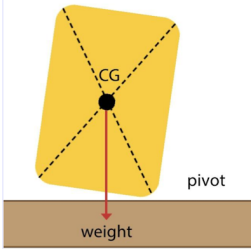
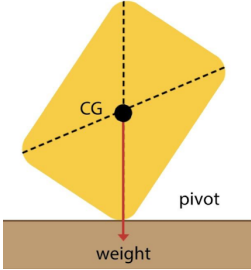
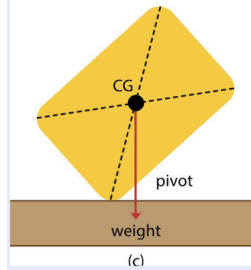


Stability

Definition: Stability of an object is a measure of its ability to return to its original position

 (a)	 (b)	 (c)
If block is tilted slightly $\Rightarrow$ it can return back to its original position. vertical line through centre of gravity still falls within its base $\Rightarrow$ Block weight creates an anticlockwise moment about the pivot that helps block return to original position	If blocks is titled more, it will reach a point where its centre of gravity is in line with the pivot and the block would balance on its edge	Block is tilted even further $\Rightarrow$ block will topple. Vertical line that passes through centre of gravity $\rightarrow$ falls outside its base. $\Rightarrow$ centre of gravity of block will generate a clockwise moment and cause block to topple

Stability of object can be affected in 2 ways $\Rightarrow$ position of centre of gravity and area of base of the object

$\rightarrow$ lower the CG $\rightarrow$ the more stable it is

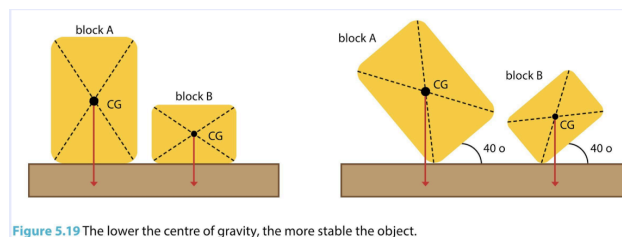


Figure 5.19 The lower the centre of gravity, the more stable the object.

same base area but of different lengths

centre of gravity of each block is indicated by CG

When tilted at same angle $\Rightarrow$ block A topples first

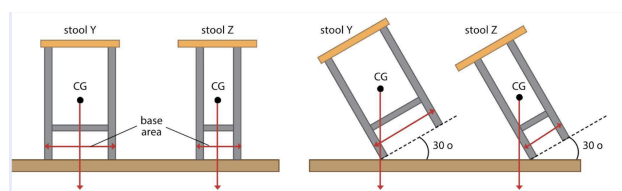
this is because block A has a higher centre of gravity

Vertical line through centre of gravity of block A falls outside its base at a smaller tilt angle

Answering Techniques for block with lower centre of gravity compared to one with higher centre of gravity

Block B has lower centre of gravity compared to block A. Block B's line of action of weight through centre of gravity falls outside its base at a much higher tilt angle compared to block A. thus, block B requires more angle of tilt to generate a clockwise resultant moment compared to A. this makes block B more stable than A, making it harder to topple

$\rightarrow$ need to check again!!



Different base area but same height

centre of gravity labelled at line CG

when slowly tilted at same angle $\Rightarrow$ Z toppled first

this is because Z has smaller base area

vertical line that passes through centre of gravity of stool Z falls outside its base area at a smaller tilt angle.

Example of toy dragonfly balancing on tip

Center of gravity is in line with the pivot (point at which tp rests on stand)

⇒ since centre of gravity (line of action of weight) is directly above pivot ⇒ no moment is generated in any given direction, hence toy does not topple

ADD ANOTHER EXAMPLE LATER